

COMP3018: Set Exercises; Human-Robot Interaction (HRI)

Cultural Differences and Probabilistic Modelling in Human-Robot Interaction

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1- Task (1): Cultural Differences and HRI Design

~1,750-word | 40% weighting | Questions 1.1-1.5

1.1. Cultural Differences in the Acceptance of Robots (Kaplan, 2004)

Kaplan's 2004 identification of East-West fundamental societal divergence is rooted in two-way observation in terms of how the culture's differences manifest as follows: **"culture affects the way technology is perceived** and, reciprocally, **technological evolution shapes culture in particular ways"** (Kaplan, 2004, p. 465); i.e. the cultural (habits), theological (religious), and mythological narrative of each region shapes the societal meta-layer (*the underlying philosophical-and-theological frameworks controlling robot integration for each culture underpinning robot acceptance*).

1.1.1 Western Society (The Frankenstein Syndrome)

Western culture has persistently viewed the creation of human-like (humanoid) entities slightly suspiciously. Kaplan (2004) identifies this as the “Frankenstein Syndrome”: a culturally-filtered conviction wherein “any artificially created humanoid will necessarily turn against its creator” (p. 475).

This anxiety traces to the West’s distinction between nature and culture, which posits “no place for hybrids” in such classifications (Kaplan, 2004, p. 470). The Western cultural narrative therefore frames humanoid robot as a challenge to human specificity (p. 476), and thus, a transgression (**TODO!!!**) against the natural order; as result, Western societies have historically envisioned robotic development towards industrial, non-anthropomorphic (instrumentalist, **TODO EXPAND**) applications wherein the machine remains a *tool* rather than a would-be social entity (**CITATION TODO**) Kaplan further notes the concept of “narcissistic shields” (*the psychological-defence mechanisms protecting human exceptionalism*) (p. 478), whereby Westerners psychologically distance themselves to manage the discomfort of encountering machines that erode the human-robot distinction.

1.1.2 Eastern Society (Technology Taming and Animism)

Japanese culture exhibits a fundamentally different ontological (*the definition of what counts as a being*) stance. Kaplan (2004) traces this to the Shinto tradition, wherein the rigid Western boundary between animate and inanimate is dissolved in favour of a “continuous network of beings” (p. 470). In this view, humanoid robots are not perceived as transgressions but instead as natural extensions.

Furthermore, Kaplan discussed the cultural mechanism of “technology taming” i.e. a recurring historical pattern wherein foreign technologies are domesticated via integration into existing cultural frameworks (p. 467). This ethos aligns with the *kata* tradition of formalised practice, where repetition leads to “maximum stability” (p. 470). The popular *Astro Boy* manga franchise (p. 466) exemplifies this domestication narratively: the robot is cast not as a Frankensteinian threat, but as a heroic companion (p. 466). Kaplan references the Amaterasu myth (p. 469) to argue that Japanese cosmology fundamentally lacks the creator-vs-creation antagonism (the inherent transgression of usurping divine privilege) that underpins Western technophobia, saying simply that “in Japan, no gods created human beings” (p. 476). This cultural openness persists into the contemporary era; as Amir explained, Japanese society remains “fine to leave people more with robots” and is “probably less sensitive to risks that might appear from robots” than the West (Lecture 5, *Ethics for AI and Robotics*).

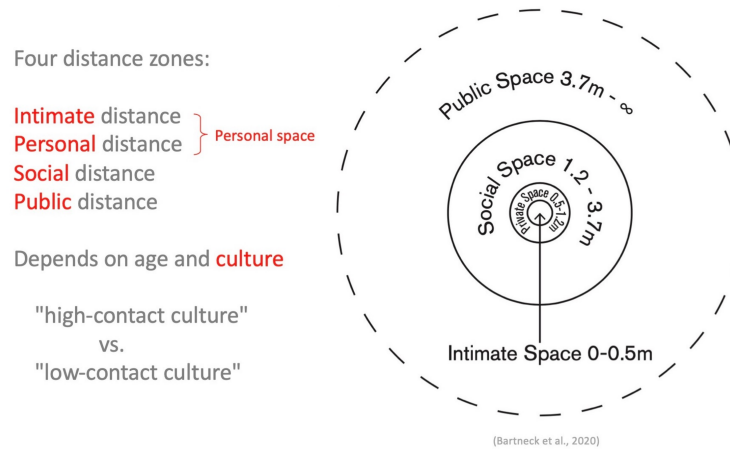
1.1.3 Implications for HRI Design.

The divergent cultural framings dictate interaction design profoundly within HRI; whilst Westerners predominantly prefer robots that maintain clear machine identity markers, thereby preserving the ‘narcissistic shield’ (Kaplan, 2004, p. 478), Eastern users instead welcome humanlike anthropomorphic features that align with animistic expectations (expectations that all entities, including artificial ones, are regarded as being a spiritual essence); Lim, Rooksby and Cross (2021, p. 1321) observed this contrasting preference, confirming that culture significantly influences the acceptance of robotic morphology as Korean participants envisioned human-like (humanoid) companion robots; US participants instead preferred machine-like functional agents.

Whilst this is true, a Western robot-designer may be inclined to impose universal proxemic standards, i.e. standards within culturally-defined personal-space boundaries based on their own cultural norms, when developing a robot. However, they’d find the approach is fundamentally flawed as people from different cultures demonstrably differ in terms of preferences with respect to proxemics (, such that what is normal for one may constitute a violation for another (Joose, Lohse and Evers, 2014, p. 1).

Regarding specific regions, Japanese normally demand larger personal-space buffers and non-tactile greeting protocols, whereas Mediterranean cultures tolerate closer approach distances (Joose, Lohse and Evers, 2014, p. 2); therefore: a culturally-calibrated model feeds local boundaries into its approach-vector calculations, as failing to respect these bounds means user expectations are contravened, effectively alienating the companion.

Proxemics



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Figure 1: Lecture-4 slide illustrating Hall's four proxemic distance zones and their cultural dependence — noting the distinction between “high-contact” and “low-contact” cultures (Bartneck et al., 2020).

Designers must also account for the novelty effect (Figure-2); the most-prominent factor: users for whom the robot represents an entirely novel stimulus exhibit inflated acceptance ratings. Despite the robot representing something novel (fictional), initial engagement masks genuine preferences, the novelty effect has *conventionally* been framed as a source of noise wherein inflated initial ratings obscure the user's authentic response (Smedegaard, 2019, p. 412). As established, the novelty effect will thus feed skewed data as it continuously inflates initial ratings, increasing the likelihood of confounding true cultural preference with transient unfamiliarity. The operative principle is cultural relativism: two things; 1) designers must use culturally-calibrated models to init interaction protocols via appropriate channels, and 2) no universal design policy can accommodate these fundamentally different ontological (real-being) commitments. Researchers cannot infer permanent acceptance from early data and instead, the system must evaluate engagement over time.

2.2.2 Novelty Effect

- Novelty effects can be particularly strong in HRI
 - Many participants may have never seen or interacted with a robot before.
 - We don't want the effects of this first interaction (so-called novelty effects) to completely overpower the manipulation of the independent variable in our study.
- One way to reduce impact:
 - Include a practice session with the robot as part of your study
 - Data from this practice session would not be recorded or used in our analysis.

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Figure 2: Lecture-6 slide defining the novelty effect in HRI and its mitigation via practice sessions.

1.2. African Cultural Factors Influencing HRI Design

Kaplan's (2004, Abstract) abstract *confines* the analysis to the East-West axis, framed explicitly as an inquiry into whether robots are "perceived in the same manner in the West and in Japan"; however, a complete account of cultural factors in HRI must address the African context, wherein distinct philosophical and socio-structural dimensions shape the robot's relational acceptance.

The following factors are well-established in African countries, and thus likely should be integrated into the design of HRI systems deployed therein:

1.2.1 Ubuntu Philosophy and Communal Identity:

Ubuntu: a Southern African philosophical principle that "a person is a person through other persons" (Metz, 2007, p. 323). Whereas Western HRI design typically foregrounds individual user experience, Ubuntu-driven design would prioritise communal benefit and relational harmony. A robot operating within an Ubuntu-oriented society should therefore be designed to address the *group* rather than the individual, facilitating collective decision-making and shared resource access (Metz, 2007, p. 324). This contrasts with the individualised personal-assistant paradigm prevalent in Western HRI.

1.2.2 Power Distance and Hierarchical Norms:

- Cirasa and Conti's (2025, p.7) 'scoping review' identifies Power Distance, i.e. a dimension of Hofstede's cultural dimensions framework (a foundational but non-peer-reviewed model just read), as a key factor shaping robot interaction norms in HRI. Addressing the literature gap they identify outside Western and East Asian spheres, it can be applied to the African context where many societies score high on this dimension, such that hierarchical authority structures strictly govern social interaction.

In regards to HRI, this suggests that robots interacting with users across different social strata must modulate behaviours accordingly: deferential language and posture when addressing elders or authority figures, and a more directive interaction style when assisting in contexts where the robot is perceived as an institutional

representative (Cirasa and Conti, 2025, p.7). Failing to encode these hierarchical norms risks contravening deeply held social expectations, thereby undermining trust.

1.2.3 Oral Tradition and Multimodal Communication:

- African cultures historically have preferred oral knowledge transmission over written documentation (Winschiers-Theophilus and Bidwell, 2013, p.21). This implicates interaction modality, i.e. voice-driven, narrative-based interfaces using speech processing and *prosodic* (rhythmically patterned) features such as pitch and MFCCs (Figure~4), may achieve higher engagement than text-heavy GUI paradigms. Furthermore, as Lecture 2 (Figure~3) established that 65% of daily-life communication is nonverbal, thus, gestural and paralinguistic channels become critical design considerations for African contexts wherein oral expressiveness is culturally normative.

**“The most important thing in communication is to hear what isn’t being said.
Peter F. Drucker**



Figure 3: Lecture-2 slide illustrating the prominence of non-verbal communication in daily life.

Affective Computing

- Speech Acoustic Features
 2. **Mel-Frequency Cepstral Coefficients (MFCC)**: describe the **overall shape of a spectral envelope**. The cepstrum is derived by taking the log of the frequency spectrum of the audio signal. Afterward, we take the cosine transform of the list of mel-log powers, then the MFCCs are the amplitudes of the resulting spectrum.
 3. **Zero Crossing Points (ZCR)**: where the waveform crosses the x-axis.
 4. **Intensity**
 5. **Duration**

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Figure 4: Lecture-3 slide listing speech acoustic features, including fundamental frequency (F0) and MFCCs.

1.2.4 Infrastructure and Access Constraints:

- Despite rapid technological growth, many African regions face infrastructure limitations including intermittent connectivity and limited access to high-specification hardware (Wyche and Steinfield, 2016). HRI systems deployed in these contexts must therefore be robust to connectivity loss, operable on low-power devices, and designed for shared rather than personal ownership — aligning with the communal ethos of Ubuntu.

1.3 Regional Design Traits (Appearance and Behaviour)

- DONE?

The cultural factors have been identified above and dictate distinct morphological and behavioural traits to maximise acceptance in the respective regions.

Figure~5 illustrates how these frameworks distinctly approach robot strategies.

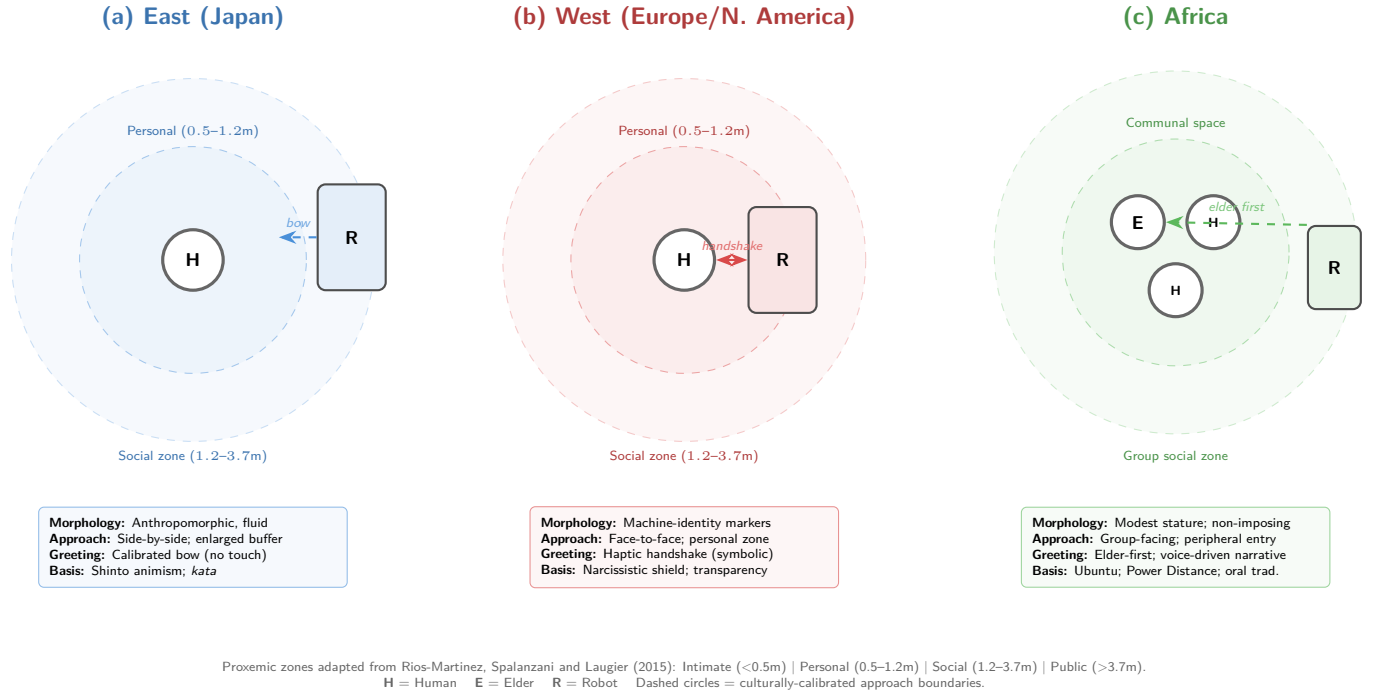


Figure 5: Culturally-adapted proxemic zones and robot approach strategies across three regional paradigms. The robot’s spatial positioning, greeting modality, and morphological design are calibrated to the cultural frameworks identified in Sections 1.1–1.2 (Rios-Martinez, Spalanzani and Laugier, 2015; Kaplan, 2004; Cirasa and Conti, 2025).

(a) The East (Japan). Because Shinto animism dissolves the natural/artificial boundary (Kaplan, 2004), anthropomorphic or highly expressive aesthetic traits are welcomed. However, to align with the *kata* tradition of harmonious form, the robot’s movements must be fluid and graceful rather than purely functional. Behaviourally, the robot should adopt a “side-by-side” cooperative posture rather than an imposing face-to-face stance, reflecting Japanese non-tactile proxemic norms requiring larger personal-space buffers (Lecture 4). Whilst this is true, a designer trained within Western conventions may be inclined to calibrate these buffers using Western proxemic data, which would systematically underestimate the spatial sensitivity of Japanese users. As Lecture 2 establishes, *haptics* (deliberate physical communication) has beneficial effects primarily within the same social group — hence physical touch is replaced by proxemic attentiveness, maintaining Hall’s personal zone (0.5–1.2m).

(b) The West (Europe/North America). To avoid triggering the Frankenstein Syndrome and to respect the “narcissistic shield” (Kaplan, 2004, p. 478), Western robots should possess functional, machine-like rather than overtly humanlike aesthetic markers (e.g., visible joints, metallic chassis) to clearly signal their artificiality, thereby preventing descent into the uncanny valley (Mori, 1970). Behaviourally, they must exhibit extreme transparency, explicitly stating their operational reasoning to alleviate fears of autonomous transgression. The handshake serves as a *symbolic gesture* (a movement with a culturally agreed-upon meaning) combined with *haptics* (Lecture 2), albeit calibrated to the Mediterranean versus Northern European proxemic distinction (Lecture 4).

(c) Africa. Informed by Ubuntu and high Power Distance (Cirasa and Conti, 2025), an African-deployed robot should possess a modest physical stature to avoid perceived challenges to human hierarchical author-

ity. Behaviourally, it must be group-facing rather than dyadic, utilising a warm, highly expressive vocal synthesiser capable of rendering the rich prosodic variations (pitch, tone) necessary for an oral-tradition society (Lecture 3; Winschiers-Theophilus and Bidwell, 2013). Furthermore, as Lecture 2 established that “65% of communication is non-verbal,” gestural and paralinguistic channels — *beat gestures* (rhythmic hand movements accentuating speech rhythm) and *iconic gestures* (movements visually representing the subject) — become critical design considerations.

1.4 Adapting Design Patterns for Sociality (Kahn et al., 2008)

Kahn et al. (2008) identify eight “psychological benchmarks” for social robots, noting the design space is “likely under-described” (p. 100). These patterns must be regionally adapted using the traits established in Section 1.3:

Pattern 1: Initial Introduction (Kahn et al., 2008, p. 100).

- (a) *East*: Rather than tactile handshakes, the robot must initiate interaction with a calibrated bow, as “in Japan it’s very considered impolite if you break the personal distance or space and try to touch somebody” (Lecture 4). The bow angle should parametrically encode social hierarchy recognition.
- (b) *West*: The *Initial Introduction* can incorporate the handshake as a tactile greeting, albeit with sensitivity to the Mediterranean (closer) versus Northern European (distant) proxemic distinctions (Lecture 4).
- (c) *Africa*: Reflecting Ubuntu’s communal orientation, the introduction must address groups rather than individuals. Encoding Power Distance norms, the robot must always greet the eldest or most-senior member first (Cirasa and Conti, 2025).

Pattern 2: Personal Interests (Kahn et al., 2008, p. 101).

- (a) *East*: The robot’s backstory can elaborately integrate into the animistic expectation of objects possessing a “spirit” or character (Kaplan, 2004).
- (b) *West*: Self-disclosure must be transparently mechanical, framing its “interests” around its programmed purpose to reinforce the user’s ontological comfort — supporting Theory of Mind mechanisms (Lecture 1) whereby the robot must infer the user’s mental state.
- (c) *Africa*: This pattern should be realised via storytelling and voice-based dialogue. The robot must establish *joint attention* (the ability of multiple agents to focus on a shared reference point) via structured eye gaze to maintain narrative authority (Lecture 2; Lecture 3).

Pattern 3: Recovering From Mistakes (Kahn et al., 2008, p. 102).

- (a) *East*: The robot should employ indirect acknowledgement strategies that preserve social harmony, avoiding direct self-criticism that may cause discomfort via loss of face.
- (b) *West*: The robot should explicitly explain its error and how it will correct it, prioritising transparency to mitigate underlying technophobia.
- (c) *Africa*: Error recovery must be strictly calibrated to social rank. An unacknowledged robot error affecting an elder constitutes a severe social violation; recovery must involve a disproportionately deferential apology to seniors, leveraging posture and movement (Lecture 2) to express contrition, thereby protecting the user’s social face within the community.

1.5 Summary of Cultural Design Implications

Table 1: Cultural factors and corresponding HRI design adaptations.

Region	Key Cultural Factor	Design Adaptation
East (Japan)	Shinto animism; non-tactile norms; <i>kata</i> tradition	Bowing protocols; enlarged personal-space buffers; indirect error recovery
West	Frankenstein Syndrome; individualism; narcissistic shields	Machine-identity markers; handshake-ready; transparent self-disclosure
Africa	Ubuntu; high Power Distance; oral tradition	Communal greetings; elder-first hierarchy; voice-driven narrative interfaces

- Kahn et al. (2008) concludes: effective HRI must be “compelling as a lived experience” (p. 104) and thus what constitutes a compelling experience is, as demonstrated above, inseparable from the cultural context thereof.

2- Task (2): POMDPs in Human-Robot Interaction

- ☒ define POMDPs
- ☐ work out word count*0.1 allowance
- ☐ FIND PEER-REVIEWED CITATIONS: POMDPs (Partially Observable Markov Decision Processes): a mathematical framework for modelling decision-making problems where the agent has incomplete information about the environment state. They extend MDPs by incorporating uncertainty in state observation, making them particularly suitable for HRI scenarios wherein the robot must infer human mental states (§trust, intent, etc) from noisy sensory data. A POMDP is defined by the tuple $\langle S, A, T, R, \Omega, O, \gamma \rangle$, where S is the set of states, A the set of actions, T the transition function, R the reward function, Ω the set of observations, O the observation function, and γ the discount factor.
- ☐ this needs to change as I am going to incorporate an AI API into the robot's system

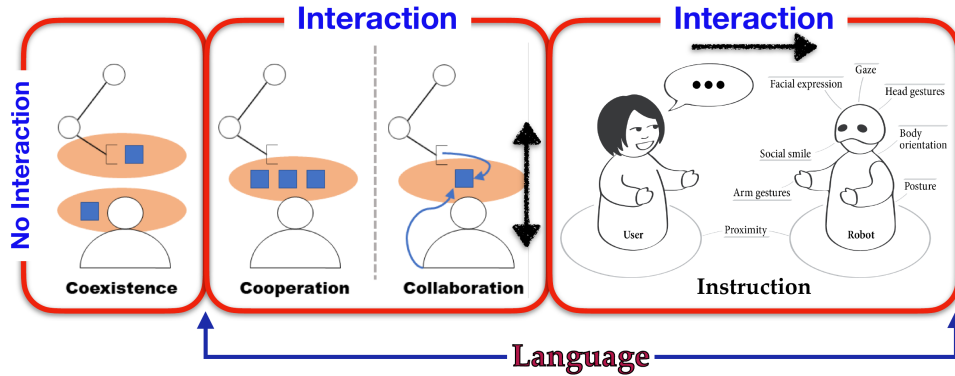
2.1 The Role of POMDPs in Trust, Cooperation, Coordination, and Collaboration

A Partially Observable Markov Decision Process (POMDP) extends the standard Markov Decision Process (MDP) — as studied in COMP3003 (Lecture 7) — by relaxing the assumption of full state observability. Whilst an MDP assumes the agent has direct access to the true environment state, a POMDP acknowledges that in most real-world HRI scenarios, the robot can only *infer* the true state via noisy, incomplete observations (Kaelbling, Littman and Cassandra, 1998). Formally, a POMDP is defined by the 7-tuple $\langle S, A, T, R, \Omega, O, \gamma \rangle$, where S is a finite set of states, A the available actions, $T(s, a, s') = P(s' | s, a)$ the transition function, $R : S \times A \rightarrow \mathbb{R}$ the reward function, Ω a finite set of observations, $O(s', a, o) = P(o | s', a)$ the observation function, and $\gamma \in [0, 1)$ the discount factor.

To understand the POMDP's utility, one must strictly differentiate interaction paradigms as defined in Lecture 1 (Figure~6). **Coexistence** involves agents sharing an environment but completing different tasks, requiring only fully-observable physical states to avoid collisions. **Cooperation** involves a shared workspace and complementary tasks. However, true **collaboration** demands a shared workspace and the *exact same shared goal* (Lecture 1). In collaboration, the robot must continuously align its actions with the human's unobservable mental states — trust, intent, cognitive load. Because the robot is fundamentally blind to these latent variables, the POMDP's belief state b becomes the computational prerequisite for graduating from mere coexistence to true collaboration. The POMDP therefore provides the principled mathematical framework for modelling trust, coordinating joint actions under uncertainty, and enabling genuinely collaborative human-robot teams (Chen et al., 2020). Indeed, Nikolaidis et al. (2017, p. 619) demonstrate this empirically via a “Bounded-Memory Adaptation Model” wherein the robot maintains a POMDP over the human's latent type and adapts its policy accordingly — showing that mutual adaptation via belief-space planning significantly outperforms fixed interaction strategies.

Human-Robot Interaction

- **Human-Robot Interaction** can be characterized in terms of interdependency by:
 - **Coexistence**: same environment & different tasks (no “or minimal” interaction)
 - **Cooperation**: same workspace & complementary parts of a task (shared goal)
 - **Collaboration**: same workspace & same task (shared goal)
 - **Instruction**: same environment (workspace) & master-slave setting



- [1] E. Matheson, R. Minto, E. G. G. Zampieri, M.Faccio and G. Rosati, “Human-Robot Collaboration in Manufacturing Applications: A Review”, *Robotics*, 8(4), 2019
- [2] J. Butepage and D. Kragic, “Human-Robot Collaboration: From Psychology to Social Robotics”, arXiv:1705.10146, 2017
- [3] B. Mutlu, “Designing Embodied Cues for Dialog with Robots”, *AI Magazine*, 32(4):17–30, 2011

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Figure 6: Lecture-1 slide defining the four HRI interaction paradigms by interdependency: coexistence, cooperation, collaboration, and instruction. The POMDP framework is necessary for graduating beyond coexistence, as collaboration requires modelling the human’s unobservable mental states.

2.2 Uncertainty, Belief States, and Decision-Making

Because the true state is hidden, the POMDP () agent maintains a **belief state** b : a probability distribution over all possible states S , where $b(s)$ represents the agent’s subjective probability that the environment is in state s , such that $\sum_{s \in S} b(s) = 1$. After taking action a and receiving observation o , the belief state is updated via **Bayesian filtering**:

$$b'(s') = \eta \cdot O(s', a, o) \sum_{s \in S} T(s, a, s') \cdot b(s)$$

η is a normalisation constant ensuring $\sum_{s'} b'(s') = 1$. This update rule captures the core epistemic challenge of HRI: the robot must continuously revise its model of the human’s internal state as new and could-be contradictory evidence arrives. As Lecture 3 discussed regarding affective computing, the robot utilises descriptors (e.g., pitch, MFCCs, zero-crossing rate) to extract observations from the human’s behaviour, and thus *feeds* these into the belief update process whereby each observation incrementally refines the robot’s understanding.

The belief state b therefore serves as a **sufficient statistic** for the entire interaction history — it compresses all past actions and observations into a single probability vector, enabling decision-making without storing the full trajectory (Kaelbling, Littman and Cassandra, 1998). This is analogous to how a social robot must infer a human’s emotional state from noisy multimodal cues (facial expressions, vocal prosody, posture) rather than accessing the “ground truth” of their feelings directly (Lecture 1; Lecture 3).

2.3 Challenges of Trust Modelling and the POMDP Response

Trust is a latent psychological variable — it cannot be directly measured, only inferred from observable behavioural indicators. Lee and See (2004, p. 54) formally define trust as “the attitude that an agent will help achieve an individual’s goals in a situation characterised by uncertainty and vulnerability” — a definition that foregrounds precisely the latent, goal-dependent nature that necessitates probabilistic modelling. Three core challenges arise. First, **the measurement problem**: observations such as task compliance rate, response latency, gaze direction (Lecture 2), and verbal affirmations are noisy proxies. Hancock et al.’s (2011, p. 520) meta-analysis of 29 empirical studies confirms this difficulty, finding that robot performance-based factors exhibit the strongest correlation with trust (mean $r = +0.26$), yet even these explain only modest variance — a user may comply with a robot’s suggestion despite low trust (e.g., due to time pressure), or indeed refuse despite high trust (e.g., due to task complexity). Second, **temporal dynamics**: trust evolves non-linearly — it builds slowly through consistent performance but degrades rapidly after errors — an asymmetry empirically confirmed by Desai et al. (2013, p. 256), who found that a single autonomous failure eroded trust more than three consecutive successes could rebuild it. The *Recovering From Mistakes* pattern (Kahn et al., 2008, p. 102) is therefore critical: a robot that acknowledges and corrects errors can arrest trust decay, whereas one that ignores failures risks appearing “aggressive” (Lecture 4). Third, **computational intractability**: solving POMDPs exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987), as the belief simplex is continuous even with finite $|S|$. Practical HRI applications therefore utilise approximate solvers — point-based methods such as PBVI (Pineau, Gordon and Thrun, 2003) or online Monte Carlo tree search such as POMCP (Silver and Veness, 2010) — to achieve tractable real-time planning.

The POMDP addresses these challenges by encoding trust as a hidden state variable, observations as probabilistic signals thereof, and actions as trust-modulating strategies (Chen et al., 2020), and thus *trust* cannot be treated as a static binary variable but must be modelled as a continuously-evolving distribution. The belief state thus provides the robot with a principled estimate of trust that is continuously refined, rather than a brittle threshold-based heuristic (a rigid binary switch wherein the subtle fluctuations of trust are ignored).

2.4 Proposed Neuro-Symbolic POMDP Model: Neo Robot with OpenAI Cognitive Architecture

To concretise this framework, I propose a neuro-symbolic POMDP model for a Neo (Pepper) humanoid robot augmented with an OpenAI multimodal API, deployed as an elderly medication-adherence assistant. The fundamental insight is this: large language models (LLMs) and vision-language models (VLMs) are powerful observation extractors — capable of parsing the once-prohibitive complexity of unstructured human behaviour into structured probabilistic assessments. However, LLMs are inherently stateless; each API call is independent, with no temporal memory of prior interactions. They are also stochastic and prone to hallucination — generating plausible but factually incorrect inferences (Ji et al., 2023). Wrapping the OpenAI API inside a POMDP belief state $b(s)$ therefore provides the mathematically rigorous temporal scaffold that the LLM alone lacks: a continuously-updated probabilistic model of the human’s latent states (Trust, Cognitive Load) that persists across the full interaction history. This neuro-symbolic paradigm — wherein a neural subsystem handles perception whilst a symbolic subsystem governs reasoning — represents what Garcez and Lamb (2023, p. 12389) term the ‘third wave’ of AI. In the robotics domain specifically, Ahn et al. (2022) demonstrated via their SayCan framework that LLMs can ground abstract language commands in physical robotic affordances, establishing the feasibility of LLM-directed action selection. However, their architecture lacks the temporal belief maintenance that a POMDP provides — a gap this model directly addresses.

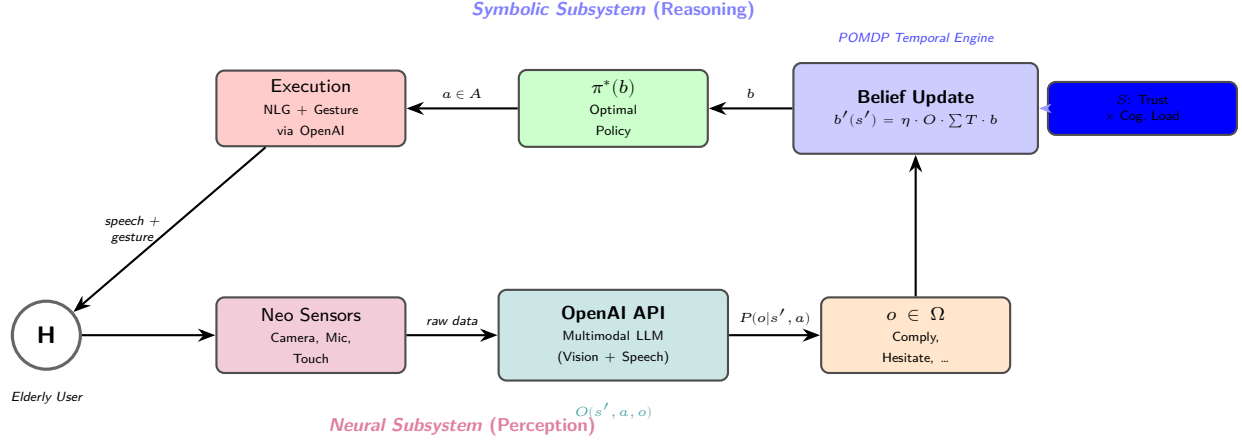


Figure 7: Neuro-symbolic architecture of the proposed OpenAI-POMDP medication-adherence system. The *neural subsystem* (bottom) utilises the OpenAI multimodal API as the observation function (O), parsing high-dimensional sensory data into discrete probabilistic observations. The *symbolic subsystem* (top) maintains the POMDP belief state and optimal policy $\pi^*(b)$, providing temporal memory via Bayesian filtering. Actions are translated back through the API into natural language and gestures for the Neo robot (Garcez and Lamb, 2023; Ahn et al., 2022).

The Neuro-Symbolic Architecture: The system operates as follows. The Neo robot’s onboard sensors (RGB camera, microphone array, tactile sensors) capture raw multimodal data from the elderly user. This data is transmitted to the OpenAI multimodal API, which serves as the **observation function** (O): the API processes the high-dimensional sensory stream — analysing facial action units via computer vision, extracting prosodic descriptors (pitch, MFCCs, zero-crossing rate) from speech (Lecture 3), and interpreting gestural semantics — to output a structured observation $o \in \Omega$. Crucially, the API provides not just a categorical label but a probability distribution over possible observations, thereby preserving the epistemic uncertainty that the POMDP requires. Within the neuro-symbolic pipeline, the observation function thus feeds structured probabilistic assessments continuously throughout the interaction process wherein the POMDP’s Bayesian update incrementally refines its model of the user.

Formal Specification:

- **State Space** (S): $\text{Trust} \in \{\text{Low}, \text{Medium}, \text{High}\} \times \text{Cognitive Load} \in \{\text{Low}, \text{High}\}$, yielding $|S| = 6$.
- **Action Space** (A): $\{\text{Verbal_Remind}, \text{Explain_Benefits}, \text{Offer_Physical_Assist}, \text{Increase_Autonomy}, \text{Disengage}\}$. The POMDP selects the abstract policy action; the OpenAI API then translates this into culturally-calibrated natural language and physical gestures executed via the Neo robot’s 25 degrees of freedom.
- **Observation Space** (Ω): $\{\text{Comply}, \text{Hesitate}, \text{Verbal_Refuse}, \text{Ignore}, \text{Gaze_Avert}\}$ — extracted by the OpenAI API from the raw multimodal stream.
- **Observation Function** (O): $P(o | s', a)$ is parametrically estimated by the API’s multimodal inference. For instance, if the true state is (High Trust, Low Cognitive Load) and the robot has performed Verbal_Remind, the API’s analysis of the user’s relaxed facial configuration and compliant vocal tone yields $P(\text{Comply} | \text{High}, \text{Low}, \text{Remind}) = 0.8$.
- **Transition Function** (T): Models trust and cognitive-load dynamics parametrically. If the true state is (Medium Trust, High Cognitive Load) and the robot performs Offer_Physical_Assist when unneeded — thereby violating personal proxemics (Lecture 4) — trust degrades: $P(\text{Low} | \text{Med}, \text{Assist}) = 0.6$, $P(\text{Med} | \text{Med}, \text{Assist}) = 0.4$. Conversely, a well-timed Explain_Benefits yields $P(\text{High} | \text{Med}, \text{Explain}) = 0.5$, $P(\text{Med} | \text{Med}, \text{Explain}) = 0.5$.
- **Reward Function** (R): Successful medication adherence yields $R = +10$; preserving user autonomy (choosing Increase_Autonomy when trust is High) yields $R = +3$; unwanted physical assistance incurs $R = -5$, reflecting the social cost of proxemic violation.

Benefits and Limitations. The model’s strength lies in its neuro-symbolic complementarity: the OpenAI API overcomes the barrier of parsing unstructured multimodal reality into discrete observations, whilst the POMDP provides the temporal cognitive engine — the belief state b — that the stateless LLM fundamentally cannot. Once initialised with a uniform prior across all six states, the belief state herein compresses all past actions and observations into a single probability vector, continuously refined via Bayesian filtering throughout the interaction wherein each observation incrementally shifts the distribution. However, the architecture introduces several limitations: 1- API latency (typically 200–800ms per inference call) may disrupt the real-time proxemic responsiveness that Lecture 4 identifies as critical; 2- the LLM’s stochastic nature means identical multimodal inputs may yield different observation distributions across calls, introducing unmodelled noise into the Bayesian update; 3- the observation model may inherit biases from the LLM’s training data — if the model was predominantly trained on younger demographics, its ability to infer emotional states from elderly facial configurations or speech patterns may be systematically degraded (Lecture 5); and 4- the discretisation of Cognitive Load into binary levels represents an information loss that future work could address via continuous-state POMDP approximations.

2.5 Ethical and Social Implications

The deployment of an LLM-augmented POMDP in HRI raises profound ethical concerns. By mathematically operationalising trust, the POMDP framework transforms a once-theoretical psychological concept into a now-computable metric. However, insofar as the reward function solely prioritises task compliance, the optimal policy may learn to *exploit* the user’s trust — timing medication requests when inferred cognitive load is highest to force compliance. Designers must therefore explicitly encode user autonomy and informed consent into the reward structure, lest assistance erode into manipulation. Sharkey (2014, p. 64) frames this via the Capability Approach: the robot must preserve the user’s capability to function independently, and thus the $R = +3$ autonomy bonus in our model is not merely a design preference but an ethical imperative — encoding the principle that dignity requires the preservation of choice.

The integration of the OpenAI API introduces three additional ethical dimensions. First, **hallucination risk**: LLMs generate plausible but factually incorrect outputs (Ji et al., 2023), and in a medication-adherence context, a hallucinated observation — misclassifying a user’s confused hesitation as willing compliance — could trigger an inappropriate action with direct health consequences. As the lecturer argued, “should the clinician trust what you are giving as judgment?” (Lecture 5); for a stochastic LLM, the answer is: not without rigorous validation, and thus Explainable AI (Lecture 5) becomes a non-negotiable requirement — the POMDP must justify *why* it selected a particular action, tracing the decision back through the belief state to the specific observations the API extracted. Second, **cloud data sovereignty**: the continuous multimodal processing of an elderly user’s facial expressions, vocal prosody, and physical behaviour — transmitted to a third-party API server — constitutes what Sharkey and Sharkey (2012, p. 35) term “surveillance by design,” now compounded by the risk of “transferring data to whichever third party we don’t verify” (Lecture 5, Figure~8). The ethical implications thereof are severe: the user’s most vulnerable moments are streamed to external servers whose data-handling policies cannot be meaningfully audited. Third, **API latency and proxemic violation**: if the observation function introduces response delays during a time-critical proxemic approach, the robot may freeze mid-interaction or fail to yield — behaviour that Lecture 4 identifies as “aggressive” (Figure~9). The ethical watchword is therefore proactive regulation (Lecture 5, Figure~10) — designers must not adopt a bottom-up approach of learning from harm after deployment, but must encode transparent, auditable constraints *ex ante* into both the reward function and the API pipeline, ensuring that the robot does not merely *model* trust but actively *earns* it through explainable, privacy-preserving behaviour.

Some Issues Raised due to the Use of Technology



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Figure 8: Lecture-5 slide raising the surveillance concern: “The AI records what you do and transfers data... to whom? Company? Third Party?” — directly applicable to the cloud-based OpenAI API pipeline proposed in Section 2.4.

Socially Appropriate Positioning

Normally, humans yield to each other when approaching

- To avoid entering each other's personal space
- Using nonverbal signals

Is this culturally dependent?

A robot that doesn't do this may appear aggressive

18

Figure 9: Lecture-4 slide establishing that a robot which fails to yield during spatial approach “may appear aggressive” — with the annotation “Is this culturally dependent?” reinforcing the cultural-calibration argument from Section 1.3.

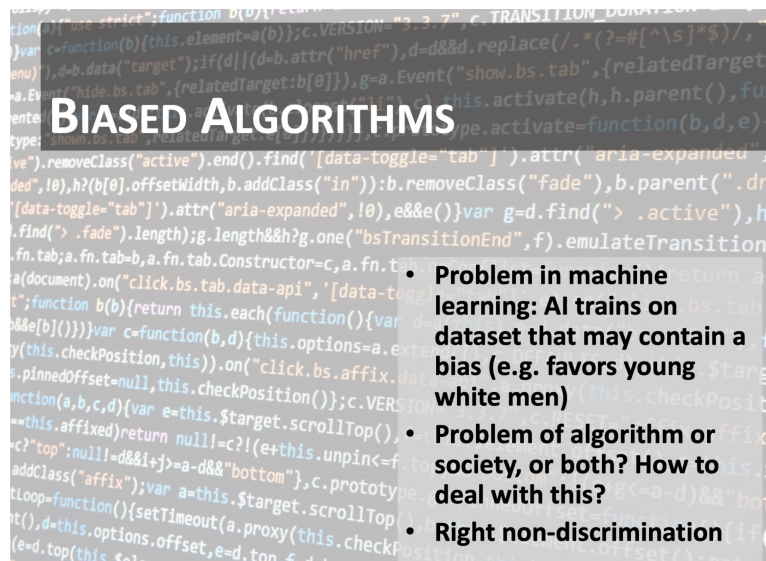
Considering Ethics in AI: Approaches



25

Figure 10: Lecture-5 slide contrasting “Bottom up” and “Pro-active” ethical approaches to AI regulation, underpinning the argument that designers must encode constraints *ex ante* rather than learning from harm after deployment.

Societal Implications of AI



19

Figure 11: Lecture-5 slide identifying the biased-algorithms problem: “AI trains on dataset that may contain a bias (e.g. favors young white men)” — relevant to the observation model’s potential demographic degradation when applied to elderly users (Section 2.4).

References

☐ PEER-REVIEWED OR CONFERENCE SOURCES ONLY

Task (1)’s

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Task (2)’s

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- ☐ make alphabetical
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Appendices

Appendix A: AI Declaration

Solo Work	S1 - Generative AI tools have not been used for this assessment.
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

Figure 12: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

AI Tool Used	Purpose of Use	Extent of Use
ChatGPT	Finding relevant pages to read in the paper (A4)	if take paper takes too long to consume efficiently
ChatGPT	General conversations via web-search AI about topic-related relates to others' studies (A4)	Few times at the end

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provide above are an accurate description of how AI was used for this assessment.